C211 - Team Project - GUI Test Cases
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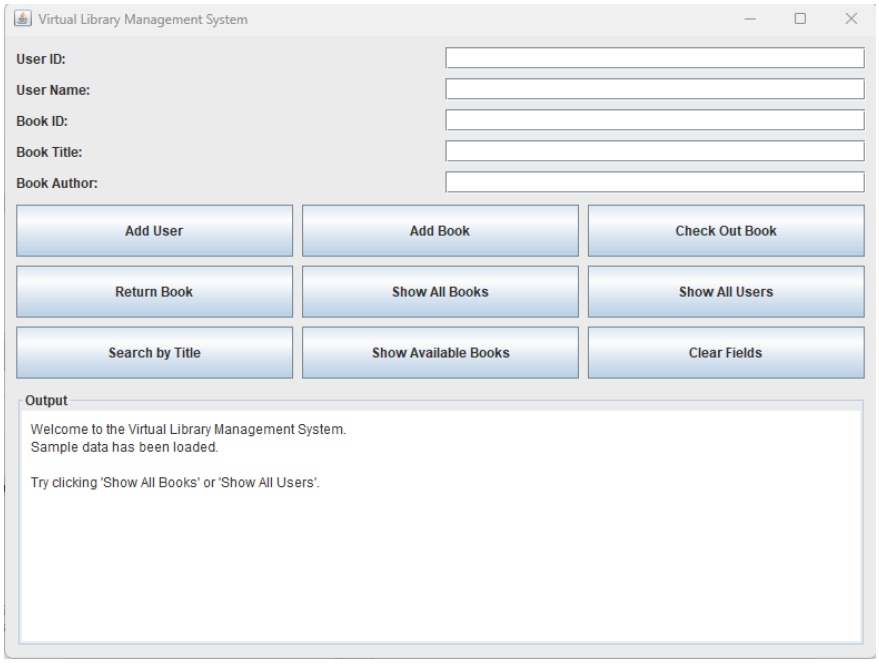
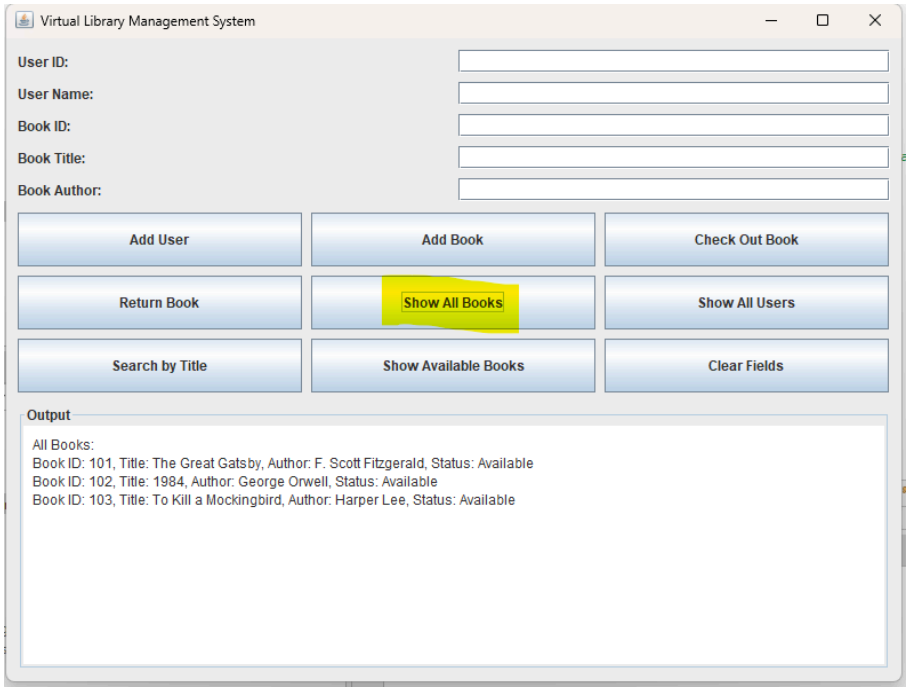
For the purpose of this report, the screenshots included represent only a sample of the test case outcomes. However, we executed the full set of functional test cases listed in this document, and all of them worked as expected.

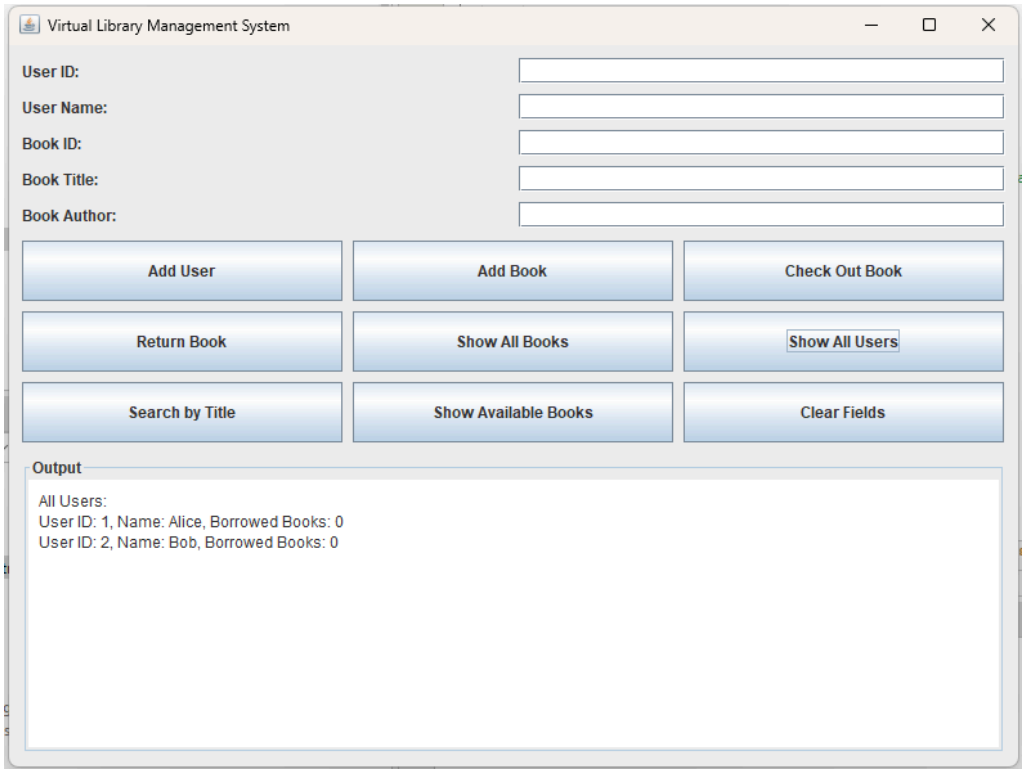
Test case 1: show GUI window (launch application)



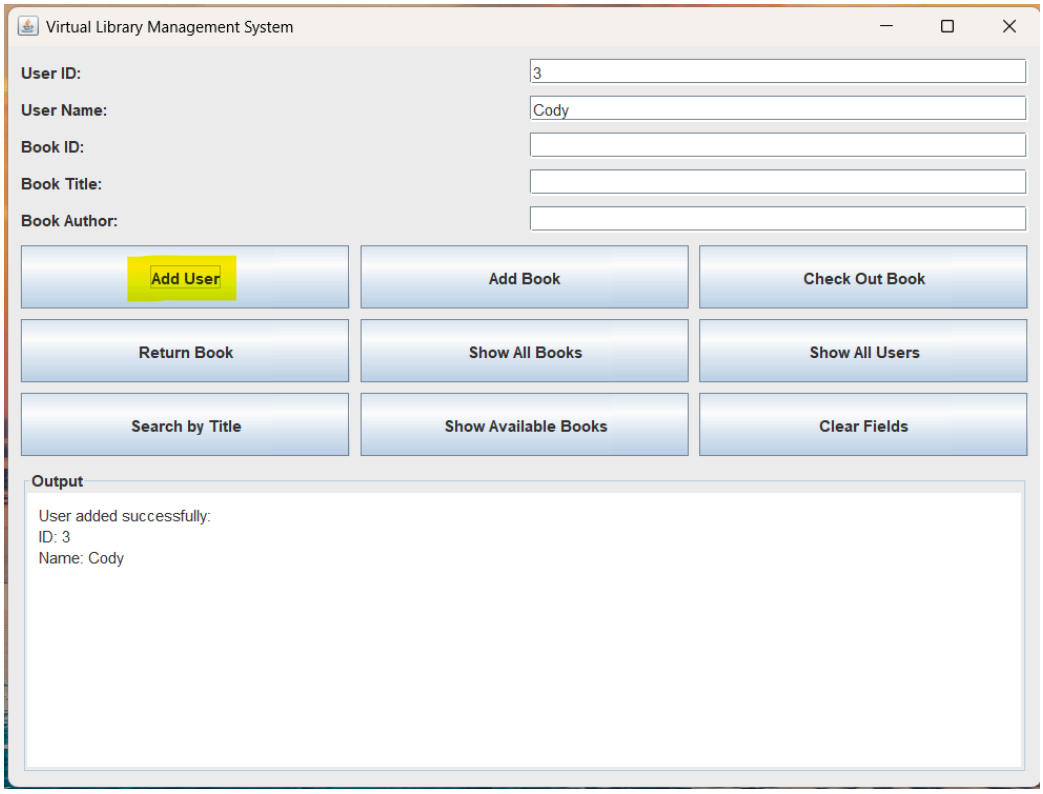
Test case 2: show all books



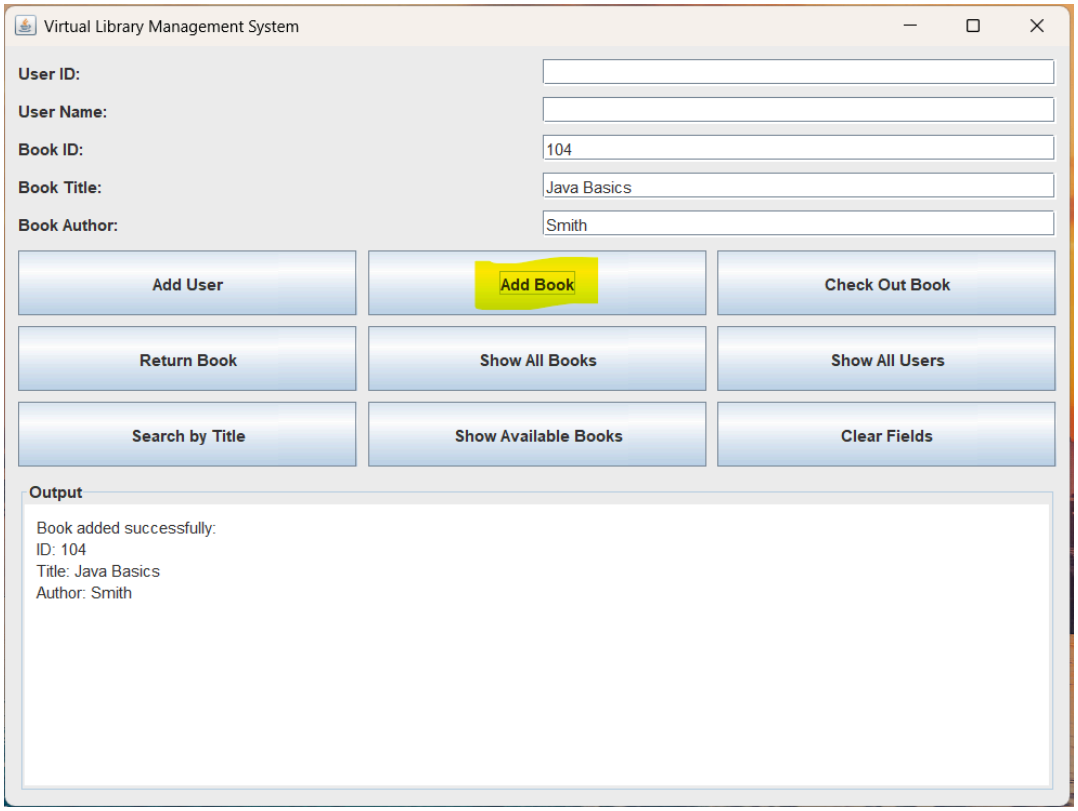
Test case 3: show all users



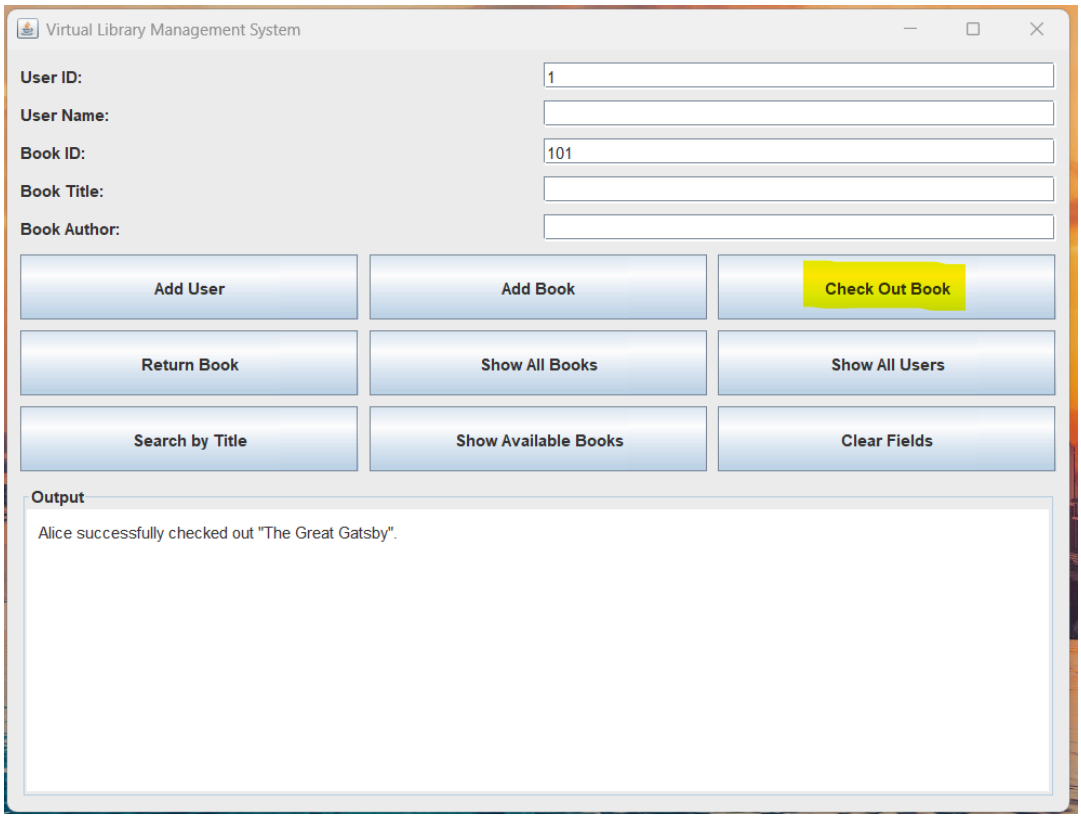
Test case 4: add user



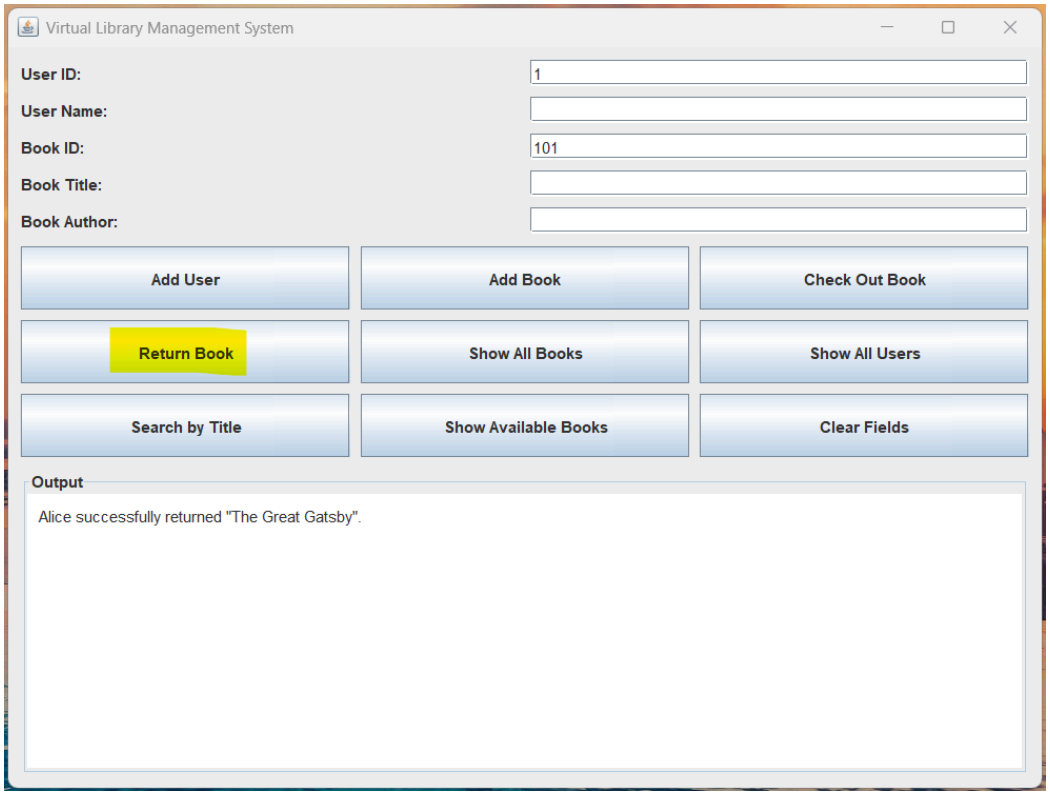
Test case 5: add book



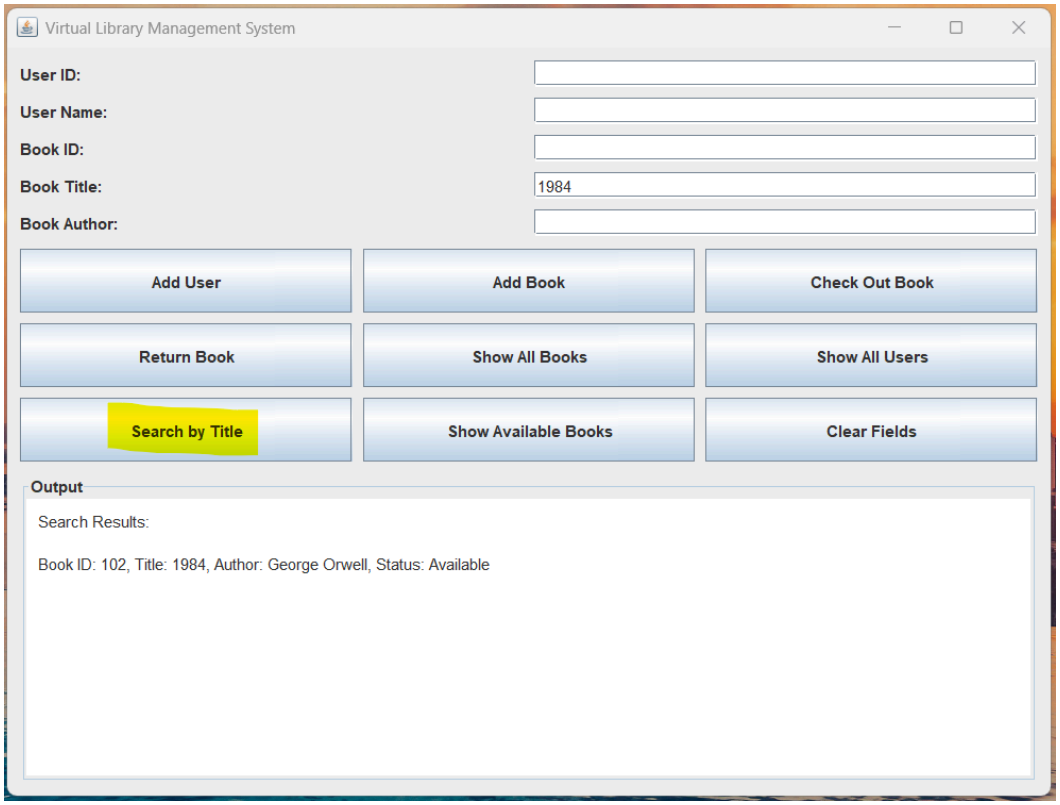
Test case 6: check out book



Test case 7: return book



Test case 8: search by title



Functional Test Cases:

#	Test Case	Input	Action	Expected Output
1	Launch application	None	Run LibraryGUI	GUI window opens with fields, buttons, and output area
2	Show all books	None	Click **Show All Books**	Displays list of sample books
3	Show all users	None	Click **Show All Users**	Displays list of users (Alice, Bob)
4	Add valid user	ID=3, Name=Cody	Click **Add User**	"User added successfully"
5	Verify added user	None	Click **Show All Users**	Charlie appears in list
6	Add user with missing name	ID=4, Name=blank	Click **Add User**	Error message prompting required fields
7	Add user with invalid ID	ID=abc	Click **Add User**	Error message for numeric ID
8	Add valid book	ID=104, Title=Java Basics, Author=Smith	Click **Add Book**	"Book added successfully"
9	Verify added book	None	Click Show All Books	Java Basics appears in list
10	Add book with missing title	ID=105, Title=blank	Click **Add Book**	Error message prompting required fields
11	Add book with invalid ID	ID=xyz	Click **Add Book**	Error message for numeric ID
12	Valid checkout	User=1, Book=101	Click **Check Out Book**	Success message
13	Book unavailable after checkout	None	Click **Show Available Books**	Book 101 not listed
14	Checkout same book twice	User=2, Book=101	Click **Check Out Book**	"Book already checked out"

15	Checkout with invalid user	User=99, Book=102	Click **Check Out Book**	"User not found"
16	Checkout with invalid book	User=1, Book=999	Click **Check Out Book**	"Book not found"
17	Borrow up to limit	User=1, Books=5 different IDs	Click **Check Out Book** repeatedly	All 5 succeed
18	Exceed borrow limit	User=1, 6th book	Click **Check Out Book**	"Maximum limit reached"
19	Valid return	User=1, Book=101	Click **Return Book**	Success message
20	Returned book available again	None	Click **Show Available Books**	Book 101 appears
21	Return book not checked out	User=1, Book=available	Click **Return Book**	"Book not checked out"
22	Wrong user return attempt	User=2, Book borrowed by User 1	Click **Return Book**	"User did not borrow book"
23	Search exact title	Title=1984	Click **Search by Title**	Matching book shown
24	Search partial title	Title=the	Click **Search by Title**	Multiple matching books
25	Search non-existent book	Title=XYZ	Click **Search by Title**	"No books found"
26	Search empty input	Title=blank	Click **Search by Title**	Prompt to enter title
27	Clear input fields	Fields filled	Click **Clear Fields**	All fields cleared